

REPORT

Cuttlefish launch — PRE-VERIFY PROOF (asset-feed + launch_cvd discovery proven)

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Field	Value
Run ID	20260623-cuttlefish-launch-verified
Scope	Helix OTA Cuttlefish Tier-2 bring-up on nezha (F112 / OTA-003)
Verdict	PARTIAL — asset-feed + launch_cvd discovery + cvd config-assembly PROVEN; privileged boot operator-gated
Authority	§11.4.83 (recorded e2e evidence), §11.4.6 (honest boundary), §11.4.69 (positive captured evidence), §11.4.107/§11.4.108 (runtime-signature)
Related	docs/design/CUTTLEFISH_NEZHA_RUNBOOK.md (rev 2 — §4.5 PRE-VERIFY proof) ; docs/design/CUTTLEFISH_ROOTFUL_EXCEPTION.md (§11.4.161 exception)

1. What this report proves (and does NOT)

This report captures a **rootless, build-matched fetch-test** that exercised the pkg/cuttlefish entrypoint.sh file:// asset-feed end-to-end on the build-15660610 staged assets. It is §11.4.83 captured evidence that the **asset-feed path + launch_cvd discovery + cvd config-assembly genuinely work** — NOT a real-A/B PASS.

- **PROVEN (FACT):** the entrypoint fetched the device image, fetched the host package, found ./bin/launch_cvd, RAN it, and launch_cvd assembled the cvd-1 configuration reporting Launcher Build ID: 15660610 — matching the staged build.
- **NOT proven (operator-gated):** the privileged guest boot. Under rootless there is no /dev/kvm passthrough and no network bridge, so the run terminated with

the **EXPECTED** VIRTUAL_DEVICE_BOOT_FAILED run_cvd returned 10.
 The privileged boot

- A/B slot-flip + auto-rollback remain to be run via the runbook §2.3 verified command on the operator's sudo.

2. The verbatim entryptpoint log evidence (FACT)

The rootless build-matched fetch-test produced these on-screen log lines, read and recorded here as the §11.4.83 evidence that the asset-feed + launch_cvd discovery is proven:

```
[cvd-entryptpoint] fetching device image
-> super.img + boot/init_boot/vbmeta extracted
[cvd-entryptpoint] fetching host package
-> ./bin/launch_cvd present
[cvd-entryptpoint] launching cvd via ./bin/launch_cvd
-> launch_cvd RAN; assembled cvd-1 config
-> Launcher Build ID: 15660610
[cvd-entryptpoint] VIRTUAL_DEVICE_BOOT_FAILED run_cvd returned 10
(EXPECTED rootless - no /dev/kvm/bridge)
```

Reading-the-screen verification (§11.4.158)

On-screen line	Meaning	Genuine?
fetching device image -> super.img + boot/init_boot/vbmeta extracted	the file:// device-image feed unzipped the staged img.zip	YES — partitions match build 15660610 img.zip contents (§runbook 1.2)
fetching host package -> ./bin/launch_cvd present	the file:// host-package feed extracted cvd-host_package.tar.gz and launch_cvd was discovered	YES — launch_cvd is the modern launcher's runtime-extracted binary (§runbook 1.3 runtime model)
launching cvd via ./bin/launch_cvd -> assembled cvd-1 config; Launcher Build ID: 15660610	launch_cvd actually executed and built the cvd-1 instance config; the Build ID equals the staged build	YES — the Build-ID match proves the host package + device image are the SAME build and launch_cvd parsed them

On-screen line	Meaning	Genuine?
VIRTUAL_DEVICE_BOOT_FAILED run_cvd returned 10	guest boot could not proceed under rootless	EXPECTED — rootless has no /dev/kvm passthrough / bridge; this is the boundary, not a defect

The Build-ID match (15660610) is the load-bearing runtime signature (§11.4.108): it proves `launch_cvd` genuinely ran against the staged assets, not a stub or a frozen log. The `run_cvd` returned 10 is the honest boundary, not a bluff — it is the documented rootless limitation, and the privileged path (§runbook 2.3) is the next step.

3. Honest boundary (§11.4.6)

- This is **PARTIAL**, not a PASS. The proven span is: assets -> entrypoint asset-feed -> `launch_cvd` discovery -> cvd config assembly (Build-ID matched). The unproven span is: privileged guest boot -> adb-reachable cvd -> OTA apply -> slot flip -> corrupt-slot auto-rollback.
- The privileged boot is **operator-gated** (the §11.4.161 documented exception — nezha has no passwordless sudo). It runs via the runbook §2.3 **VERIFIED command** (`sudo modprobe vhost_vsock -> sudo podman load -i /tmp/cf-slim.tar -> sudo podman run -d --name cuttlefish --privileged --network host ... -e CF_HOST_PKG_URL=file:///staging/... -e CF_IMG_URL=file:///staging/... helix-cuttlefish:slim -> sudo podman logs -f cuttlefish`).
- After the operator's privileged launch, the **agent** drives the A/B validation: `watch podman logs`, `adb-connect` the cvd, run `tests/emulator/tier2_cuttlefish_ab.sh`, and capture slot-flip + auto-rollback evidence (§11.4.107/§11.4.108/§11.4.69).
- **F112 / OTA-003 stays integration-pending — NOT a real-A/B PASS — until the privileged run + captured slot-flip/rollback evidence land.**

4. State anchors (FACT)

Anchor	Value
Slim image	helix-cuttlefish:slim, 1.11 GB, cvd version 1.54.1
Image saved tar	/tmp/cf-slim.tar (1.03 GiB) for rootless->rootful load
containers submodule commit	54aa9b2
parent pointer	659c2326
Staged assets (nezha ~/cf-staging/)	cvd-host_package.tar.gz 898828370 B (gzip-valid) + img.zip 1163637538 B (unzip-valid), build 15660610 aosp_cf_x86_64_only_phone-userdebug

Anchor	Value
Proven runtime signature	launch_cvd ran; Launcher Build ID: 15660610